

NAV-BOT

Status

This document defines NAV-BOT's core concept, theory of change, and MVP scope. It is a living reference, not a finalized implementation.

What is NAV-BOT?

NAV-BOT helps students build healthier cognitive defaults over time by interrupting maladaptive coping patterns and guiding them toward more reflective, prefrontal-based responses when stress arises.

NAV-BOT helps young adults practice healthier ways of responding to stress so those responses become easier and more automatic over time.

NAV-BOT reduces how often stress teaches the brain that avoidance, self-attack, or shutdown are the safest options.

1. Base Layer (Always On): Safe Venting Space

This is the foundation, not a side feature.

Design principles

- Bright, Gen-Z-friendly UI
- Sentiment-colored message bubbles
- GPT + Rasa for conversation and intent / pattern detection
- No pressure to "do" anything

User behavior

- Users can talk freely
- Dump thoughts
- Journal
- Reflect

Value this layer provides

- Emotional containment
- Self-expression
- Pattern visibility over time

This makes NAV-BOT safe and approachable — not a productivity cop.

2. Pattern Awareness (Consent-Based)

Patterns appear naturally in chat.

NAV-BOT listens for:

- Avoidance
- Catastrophizing
- Rumination
- Overwhelm
- Shutdown / distraction language

NAV-BOT does **not** interrupt randomly.

It intervenes **only when a recognizable pattern appears**.

Consent-first interruption

NAV-BOT might say:

"I'm noticing a pattern that often shows up when people feel overwhelmed.
Do you want help working through it, or do you want to just keep venting?"

Why this matters

- The user stays in control
- No therapy overreach
- Minimal resistance

Once consent is given, NAV-BOT selects the appropriate tool.

What University Students Actually Struggle With (Problem Space)

A. Academic Pressure

- Procrastination
- Imposter syndrome
- Test anxiety
- Burnout
- Perfectionism
- Overwhelm from deadlines
- Fear of failure

B. Emotional Health

- Anxiety (general + social)
- Depression symptoms
- Loneliness
- Rumination
- Negative self-talk
- Unstable routines

C. Executive Function (Brain-Based)

Common challenges among Gen-Z students:

- Working memory overload
- Task switching fatigue
- Attention dysregulation
- Planning difficulty
- Reward-system dependency for motivation

Often overlaps with ADHD tendencies (including undiagnosed).

D. Identity & Social Stress

- Balancing independence with academic pressure
- Relationships
- Rejection sensitivity
- Homesickness
- Lack of life structure

3. CBT Tool Kit (MVP Core)

If the user opts in, NAV-BOT chooses **one of three tools**.

[1] Safe Venting (Stay Here)

Used when

- Emotional load is high

What NAV-BOT does

- Reflective responses
 - Validation
 - Sentiment-based visual feedback
 - No fixing
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[2] CBT Thought Challenge (Brain-Based Explanations)

Used when

- Distorted thinking appears

Flow

- Identify the cognitive distortion
- Ask 1–2 grounding questions
- Offer a balanced reframe

Short. Bounded. Optional.

Example responses

- “Your brain is in survival mode, not failure mode.”
- “This is your amygdala being louder than your prefrontal cortex.”
- “Lack of motivation isn’t a character flaw — it’s dopamine dysregulation.”

These explanations reduce shame and build awareness.

3 Study / Unstuck Mode (Task Breakdown)

Used when

- Avoidance or overwhelm appears

This is where task breakdown lives.

What NAV-BOT does

- Lowers perceived threat
- Reduces cognitive load
- Shifts from evaluation → action

How

- Breaks tasks into very small chunks
- Offers time-block suggestions
- Reduces mental load
- Uses compassionate accountability

UI behavior

- Converts vague stress into a small, concrete checklist
- Steps are intentionally tiny and time-bounded
- Users can check items off as they go

Example steps

- ☐ Stand up / get water (2 min)
- ☐ Open notes (no reading yet)
- ☐ List topics only (5–10 min)

Design rules

- Finishing everything is *not* required
- Stopping on purpose is allowed
- Progress = approach, not completion

Optional (with consent)

- Soft focus mode
- Timers or reminders
- Easy exit at any time

This is behavioral activation — not productivity pressure.

4. Awareness Layer (Optional, Brief)

Only **after** the user is calmer or moving.

Purpose

- Reduce shame
- Build metacognition
- Explain *why* something helped

Example responses

- "Breaking this down helped your planning brain come back online."
- "Your brain is in survival mode, not failure mode."
- "This is your amygdala being louder than your prefrontal cortex."

Never before action. Never as a lecture.

Neuroscience Behind the CBT Approach (Context)

Thoughts → Emotions → Behaviors

Thoughts activate:

- Amygdala (emotion / threat)
- Prefrontal cortex (planning / reasoning)

CBT supports prefrontal-driven regulation during stress.

Cognitive Distortions = Autopilot Errors

Common distortions:

- Catastrophizing
- Mind-reading
- All-or-nothing thinking

These activate threat circuits.

CBT teaches identification and challenge.

Behavioral Activation

Low mood → less activity → lower mood.

CBT reverses this by acting first and letting mood follow.

Exposure & Reward Prediction

- Test anxiety reduces with controlled exposure
- Dopamine loops (notifications, scrolling) distort motivation
- CBT reframes rewards toward long-term motivation

5. Analytics Dashboard (Later / MVP+)

- Mood tracking + sentiment graph
 - Gen-Z-styled dashboard
 - Weekly mood trends
 - "Emotional weather report"
 - Stress spike highlights
 - Gentle predictions (e.g., "Mondays seem hardest for you")
 - Personal insights
 - AI-generated summaries
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Long-Term Effect: Supporting Resilient Coping Patterns

1 Early pattern recognition

- "This looks like catastrophizing."
- "This sounds like avoidance."
- "This feels like rumination."

2 Intervention at the right moment

- Not hours later
- Not in a lesson
- When the pattern appears in chat

3 Guided alternative responses Instead of:

- Avoidance → scrolling
- Rumination → spiraling
- Threat → shutdown

NAV-BOT nudges toward:

- Task shrinking
- Reframing
- Values-based action
- Bounded effort

This is prefrontal practice, not advice.

4 Over time

- Interruptions happen earlier
- Avoidance windows shrink
- Users trust themselves more
- Healthier responses become defaults